PERFECT WIRELESS EXPERIENCE

# Firmware Upgrade Guide_Linux

V1.9

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# Applicable Models

| No. | Applicable Model | Description |
|---|---|---|
| 1 | L716/L718 | ZTE V3E/T |
| 2 | L61x/LC61x/MC61x/LG61x | UNISOC 8910 |
| 3 | MC66x/MG66x | UNISOC 8850 |
| 4 | FG650/FG652/FM650 | UNISOC UDX710 |
| 5 | NL668/MC116/LC116 | QCOM MDM9x07 |
| 6 | MA510/MC109/MC100E | QCOM MDM9205 |
| 7 | FG10x/FM10x | QCOM X12 |
| 8 | FM150/FG150 | QCOM X55 |
| 9 | FM160/FG160 | QCOM X62 |
| 10 | FG621 | UNISOC SL8563 |
| 11 | LE270 | EC718 |
| 12 | FG180/FM180 | QCOM X72 |
| 13 | FG190/FM190 | QCOM X75 |
| 14 | FM550/FG550 | S5300 |

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# Change History

| | |
|---|---|
| V1.9 (2025-02-22) | Added the Samsung (S5300) platform upgrade function |
| V1.8 (2024-11-29) | Added the description of remote upgrade |
| V1.7 (2024-03-11) | Add the usage description of the -c parameter |
| V1.6 (2024-03-01) | Add applicable model SDX72 |
| V1.5 (2023-11-23) | 1.Modify the FG621 chip signal<br>2. Added core shifting platform and related description of the platform |
| V1.4 (2023-11-07) | Improved the tool adaptation model and -r1 parameter description |
| V1.3 (2023-10-23) | The power-off message cannot be displayed during the upgrade |
| V1.2 (2023-08-23) | Modifying Parameters |
| V1.1 (2023-05-20) | Added the description of upgrading PCIE ports on a Qualcomm platform |
| V1.0 (2022-08-28) | Initial version. |

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# 1 Introduction

Upgrade_ tool is used for Linux and Android hosts to upgrade the firmware of Fibocom modules.

# 2 Compilation

## 2.1 Upgrade Tool Package Introduction

- doc: Chinese and English guidance documents
- misc_code: common code
- qcom_code: Qcom modules upgrade code
- zte_code: ZTE modules upgrade code
- unisoc_code: Unisco modules upgrade code
- eigencomm _code: eigencomm modules upgrade code
- smr5300_code: samsung modules upgrade code
- main.c: The program code
- Makefile: Compiling configuration file for Linux Environment
- Android.mk: Compiling configuration file for Android Environment

The detailed directory structure is as shown in the figure 1.

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```
├── Android.mk
├── doc
│   ├── Fibocom_Linux_Firmware_Upgrade_Guide.pdf
│   └── Fibocom_Linux固件升级指南.pdf
├── main.c
├── Makefile
├── misc_code
│   ├── misc.c
│   ├── misc.h
│   ├── misc_usb.c
│   ├── misc_usb.h
│   ├── usb2tcp.c
│   └── usb2tcp.h
├── qcom_code
│   ├── firehose_download.c
│   ├── firehose_download.h
│   ├── md5sum.c
│   ├── md5sum.h
│   ├── pcie_download.c
│   ├── pcie_download.h
│   ├── qcom_devices_list.h
│   ├── qcom_main.c
│   ├── sahara_protocol.c
│   ├── sahara_protocol.h
│   ├── stream_download.c
│   └── stream_download.h
├── README
├── unisoc_code
│   ├── crc.c
│   ├── nv.c
│   ├── pac.c
│   ├── pac.h
│   ├── unisoc_devices_list.h
│   ├── unisoc_download.c
│   ├── unisoc_download.h
│   ├── unisoc_main.c
│   └── xml.c
└── zte_code
    ├── zte_devices_list.h
    ├── zte_download.c
    ├── zte_download.h
    └── zte_main.c

5 directories, 37 files
```

Figure 1. Directory structure

## 2.2 Compile Upgrade Tool

### 2.2.1 Linux Upgrade Tool

- Configure cross compilation tool

GCC is used by default. If arm GCC is required, set CROSS_COMPILE variable in Makefile.

As blow:

```
# If use the arm gcc, please set the arm gcc path.
# ARM_GCC = ~/ARM_Linux_GCC/bin/arm-none-linux-gnueabi-

CC = $(ARM_GCC)gcc

CFLAGS = -g -Wall

LD_LIBRARY = -lpthread

INCLUDE = -I./zte_code -I./qcom_code -I./unisoc_code -I./misc_code
SOURCES = $(wildcard ./zte_code/*.c ./qcom_code/*.c ./unisoc_code/*.c ./misc_cod
e/*.c ./*.c)

BIN_TARGET = upgrade_tool

linux: clean
        ${CC} ${CFLAGS} -s ${SOURCES} ${INCLUDE} -o ${BIN_TARGET} $(LD_LIBRARY)

.PHONY: clean

clean:
        @rm -rvf $(BIN_TARGET)
~
~
~
"Makefile" [dos] 22L, 515C                                  22,0-1        All
```

Figure 2. Makefile

- Compilation

Put the upgrade tool code on the Linux host, then in the code directory, execute make. The upgrade_tool will be generated if the compilation is OK.

As shown in the following figure.

```
ght@ght-pc:~/Fibocom_Linux_Firmware_Upgrade_V1.1.0.0
$ ls
Android.mk  main.c    misc_code  README       zte_code
doc         Makefile  qcom_code  unisoc_code
ght@ght-pc:~/Fibocom_Linux_Firmware_Upgrade_V1.1.0.0
$ make
gcc -g -Wall -s ./zte_code/zte_download.c ./zte_code/zte_main.c ./qcom_code/saha
ra_protocol.c ./qcom_code/firehose_download.c ./qcom_code/pcie_download.c ./qcom
_code/stream_download.c ./qcom_code/qcom_main.c ./qcom_code/md5sum.c ./unisoc_co
de/xml.c ./unisoc_code/unisoc_download.c ./unisoc_code/crc.c ./unisoc_code/uniso
c_main.c ./unisoc_code/nv.c ./unisoc_code/pac.c ./misc_code/usb2tcp.c ./misc_cod
e/misc_usb.c ./misc_code/misc.c ./main.c -I./zte_code -I./qcom_code -I./unisoc_c
ode -I./misc_code -o upgrade_tool -lpthread
ght@ght-pc:~/Fibocom_Linux_Firmware_Upgrade_V1.1.0.0
$ ls
Android.mk  main.c    misc_code  README       upgrade_tool
doc         Makefile  qcom_code  unisoc_code  zte_code
ght@ght-pc:~/Fibocom_Linux_Firmware_Upgrade_V1.1.0.0
$
```

Figure 3. Compile generated result

The eigencomm platform currently only supports 32-bit compilation, which includes - m32 in the compilation option. If an error is reported during the compilation process: sys/cdefs.h: No such file or directory, it indicates that the Linux host needs to install 32-bit library files. In Ubuntu, the sudo apt-get install libc6-dev-i386 command to install online.

## 2.2.2 Android Upgrade Tool

1. Put the upgrade tool code into the Android code directory.
2. Run source build/envsetup.sh.
3. Run lunch, and then select the build option.
4. Run mmm Fibocom_Linux_Firmware_Upgrade_XXX
5. If the compilation is successful, the upgrade_tool will be generated

The path of upgrade_tool will be displayed in the compilation log.

e.g.

out/target/product/msm8953_64/system/bin/upgrade_tool

> Android system tools for Android 8 and higher are not currently compatible.

```
ght@ubuntu-230:~/android_work
$ source build/envsetup.sh
======BSP_SELECT_ID=======
including device/qcom/common/vendorsetup.sh
including vendor/qcom/proprietary/common/vendorsetup.sh
including sdk/bash_completion/adb.bash
ght@ubuntu-230:~/android_work
$ lunch msm8953_64-user

============================================
PLATFORM_VERSION_CODENAME=REL
PLATFORM_VERSION=7.1.2
TARGET_PRODUCT=msm8953_64
TARGET_BUILD_VARIANT=user
TARGET_BUILD_TYPE=release
TARGET_BUILD_APPS=
TARGET_ARCH=arm64
TARGET_ARCH_VARIANT=armv8-a
TARGET_CPU_VARIANT=generic
TARGET_2ND_ARCH=arm
TARGET_2ND_ARCH_VARIANT=armv7-a-neon
TARGET_2ND_CPU_VARIANT=cortex-a53
HOST_ARCH=x86_64
HOST_2ND_ARCH=x86
HOST_OS=linux
HOST_OS_EXTRA=Linux-4.4.0-31-generic-x86_64-with-Ubuntu-14.04-trusty
HOST_CROSS_OS=windows
HOST_CROSS_ARCH=x86
HOST_CROSS_2ND_ARCH=x86_64
HOST_BUILD_TYPE=release
BUILD_ID=N2G47H
OUT_DIR=out
============================================
ght@ubuntu-230:~/android_work
$ mmm Fibocom_Linux_Firmware_Upgrade_V1.1.0.0
============================================
PRODUCT_COPY_FILES device/qcom/common/media/media_profiles.xml:system/etc/media_profiles.xml ignored.
PRODUCT_COPY_FILES device/qcom/common/media/media_codecs.xml:system/etc/media_codecs.xml ignored.
No private recovery resources for TARGET_DEVICE msm8953_64
616+0 records in
616+0 records out
630784 bytes (631 kB) copied, 0.000854025 s, 739 MB/s
Starting build with ninja
ninja: Entering directory `.'
[100% 25/25] Install: out/target/product/msm8953_64/system/bin/upgrade_tool
make: Leaving directory `/fwork1/zhoumaiyun/android_work'

#### make completed successfully (7 seconds) ####
```

Figure 4. Android environment compilation procedure

# 3 Upgrade

## 3.1 Local Upgrade

1. Check whether the USB connection of the module is normal.

> Qcom module:
>
> 1. If the qcom module is in normal mode, please send at+disk=0,0,0 to unlock diag port first.
> 2. During the upgrade, ensure that the module cannot be powered off.
>
> Eigencomm module:
>
> 1. At present, it only supports pressing the boot button when powered on, and the module enters forced download mode, using USB for firmware upgrade.
> 2. The upgrade process relies on files in the config folder, so upgrade programs in the same level directory must exist in the config folder.

2. Copy the firmware file and upgrade_tool to the host.
3. Enter the directory where the upgrade_tool is located, and then execute the upgrade command.

```
./upgrade_tool -f firmware image or dir
```

e.g.

```
qcom:    ./upgrade_tool -f 19010.1000.00.02.73.15/Maincode
unisoc:   ./upgrade_tool   -f   16000.1000.00.06.01.05.pac
zte:            ./upgrade_tool   -f   17016.1000.00.38.01.21.bin
eigencomm：./upgrade_tool   -f   12007.6000.00.02.02.07-718.binpkg
samsung：./upgrade_tool   -f   83600.1000.00.01.01.01
```

## 3.2 Remote upgrade

If the upper computer does not have enough space to store the module firmware package, upgrade_tool can upgrade the firmware over the network using a remote PC.

The steps are as follows:

1. Place upgrade_tool on the host.
2. Place the firmware package and upgrade_tool on the remote PC.
3. Run upgrae_tool on the host to start the TCP service.

```
./upgrade_tool  -p 9008
```

4. View the IP address of the host. You can ping the IP address from a remote PC.
5. Run the download command on the remote PC.

```
qcom:   ./upgrade_tool   -p IP:9008  -f  19010.1000.00.02.73.15/Maincode –t 2
```

```
unisoc:   ./upgrade_tool  -p IP:9008  -f  16000.1000.00.06.01.05.pac –t 3
```

1. Only Qualcomm platform/Unisoc platform supports remote upgrade
2. Because the remote PC is not connected to a module, the specific platform of the module cannot be distinguished. You need to specify the -t parameter to determine the platform

## 3.3 Viewing Upgrade Result

- The following log will be printed after the upgrade succeeds:

```
Upgrade module successfully
```

- If the upgrade fails, you can get the reason for the upgrade failure from the download log.
- If you run the download tool with the - l parameter, the download tool will save the upgrade log file to the corresponding directory, with the log file name fibo_download_xxx.log.

# 4 Upgrade Parameter Description

| No. | Parameter | Necessary | Description |
|---|---|---|---|
| 1 | -f firmware image or dir | Yes | Upgrade firmware image dir |
| 2 | -l <log dir> | No | set log dir and save the upgrade log. |
| 3 | -r 0/1 | No | This parameter is only applicable to MDM9x07 and MDM9205 platform. When -r 1 is set, NV will be restored automatically after upgrade. If -r 1 is not set, you need to manually execute at command to restore NV after upgrade |
| 4 | -d <port> | No | Download port (/dev/ttyUSBX) |
| 5 | -z <0/1> | No | Send Zero-length package. It is 0 by default. No special instructions, no need to set parameters. |
| 6 | -e | No | Erase ALL partitions before upgrading. Defaults is 0. No special instructions, no need to set parameters. |
| 7 | -c | No | Specifies the xml path to use for the qcom platform upgrade |
| 8 | -p <IP:9008/9008> | No | Remote upgrade parameters:<br>Firmware package side: -p IP:9008<br>Module side: -p 9008 |
| 9 | -t <1/2/3/4/5> | No | Designated platform<br>If the system is connected to multiple platform modules, use -t to specify:<br>-t 1 indicates ZTE.<br>-t 2 indicates Qualcomm.<br>-t 3 indicates UNISOC.<br>-t 4 indicates Eigencomm.<br>-t 5 indicates Samsung. |

1.The PCIE port on the Qualcomm platform needs to be upgraded using -d /dev/mhi. Otherwise, the upgrade cannot be performed.

2.The eigencomm platform currently only supports the - f parameter

3. If FG131 series products need full erase upgrade (using the -e parameter), you need to use the -c parameter to specify the xml with erase in the firmware package, as shown in the following example:

```
/upgrade_tool -e -f 89170.1000.00.01.02.06 /Maincode -c
89170.1000.00.01.02.06                                    /Maincode/
rawprogram_nand_p4K_b256K_for_erase_dl.xml
```

# 5 NV Restoration Commands

## 5.1 NV Restoration of MDM9X07 and MDM9205 Modules

- Execute upgrade_tool with the -r 1 parameter, the NV can be restored automatically when the module is powered on for the first time.

e.g.

```
./upgrade_tool  -f  19010.1000.00.02.73.15/Maincode  -r 1
```

- If the parameter -r 1 is not set when upgrading firmware, you can execute the at command to restore NV.

The operation steps:

1. Send at command: at+efserrfatal
2. If the module does not restart automatically, please send reboot command: at+cfun=15
3. The module will restart and auto restore NV.
4. Wait until the module is started, and check whether IMEI and SN is OK.

MDM9205 TX modules cannot upgrade EFS. If static NV in the EFS need to update, you need to use Windows Tool Fibocom_ MDM_ Multiupdater to upgrade.

## 5.2 NV Restoration of Other Modules

There is no need to add the -r parameter when downloading. The module will automatically restore NV after downloading.